

4(a)(i) Since g and r are constant, so a is proportional to x .
Negative sign shows that a and x are in opposite direction. [APPLICATION]
Hence the ball undergoes simple harmonic motion [CONCLUSION]

(ii) $\omega^2 = \frac{g}{r}$ and $\omega = \frac{2\pi}{T}$ [CONCEPT]
 $\omega^2 = \frac{9.81}{0.28} = 35$

$$T = 1.06 \text{ s}$$

$$\tau = 0.53 \text{ s}$$

- (b)** Sketch:
time period constant (or increases very slightly)
drawn lines always 'inside' given loops, up to given time duration
successive decrease in peak height

5(a)(i) Faraday's law states that the emf induced in a conductor is proportional to the rate of